

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter (for API load testing)
Type of Testing	Load Testing + Stress Testing
Target Module / API	predict - approval endpoint (Flask API)
Test Environment	Hosted on Render (Free Tier)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	Virtual Users	Duration (sec)	Expected Outcome
1	10 users submitting applications simultaneously	10	60	Avg response < 2s
2	Spike of 50 users at once	50	30	System handles spike without crashing
3	Continuous load of 20 users	20	120	Stable throughput, <1% error rate
4	Stress test with 100 users	100	45	Graceful degradation, no server crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2s	1.6s	Pass	Within acceptable range
2	Response Time (Max)	< 5s	4.2s	Pass	Slight spike under stress
3	Throughput (Req/sec)	≥ 20	28	Pass	Good scalability
4	Error Rate	< 1%	0.5%	Pass	Minor errors under heavy load
5	CPU Utilization	< 80%	72%	Pass	Stable
6	Memory Utilization	< 80%	65%	Pass	No leaks observed

Step 4: Observations & Analysis

- **Key Findings:** The API consistently responded under 2 seconds for normal loads. Stress testing showed resilience up to 100 users.
- **Bottlenecks Identified:** Slight increase in response time during spikes.
- **Optimization Steps Taken:** Added caching for repeated requests, optimized database queries, and enabled gzip compression for faster responses.

Step 5: Screenshots / Evidence



Credit Card Approval Predictor

Submit an application to receive an instant, model-driven approval decision.

Model cc-score-v1.3.0

API connected · key verified · predict-approval

API reachable · key verified · validation active

Recheck

Applicant details

Fill sampleReset

FULL NAME

hina kumar

JOB TITLE

assistant manager

AGE

26

ANNUAL INCOME

\$ 45000

Whole years

CREDIT SCORE

448

FICO-style

EMPLOYMENT YEARS

6

USD, before taxes

Time in current role

DEBT-TO-INCOME RATIO

% 0.5

Monthly debt payments divided by gross monthly income

MARITAL STATUS

Single

GENDER

Female

INDUSTRY

Technology

Predict

DECISION

 Declined

RISK BAND

Moderate Risk

Approval probability

45%

SCORE

45/100

LATENCY

1801 ms

MODEL

cc-score-v1.3.0

TOP CONTRIBUTING FACTORS

Debt-to-Income	-1.10
Annual Income	+0.93
Credit Score	+0.70
Employment History	+0.18
Industry	+0.16

Prediction for hina kumar - logged to audit trail

This applicant falls below the model's approval threshold.
Recommend manual review.